

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement an intelligent traffic control system using finite state machine methodology.	
Experiment No: 10	Date:	Enrolment No: 923017330313

Aim: To design and implement an intelligent traffic control system using finite state machine methodology.

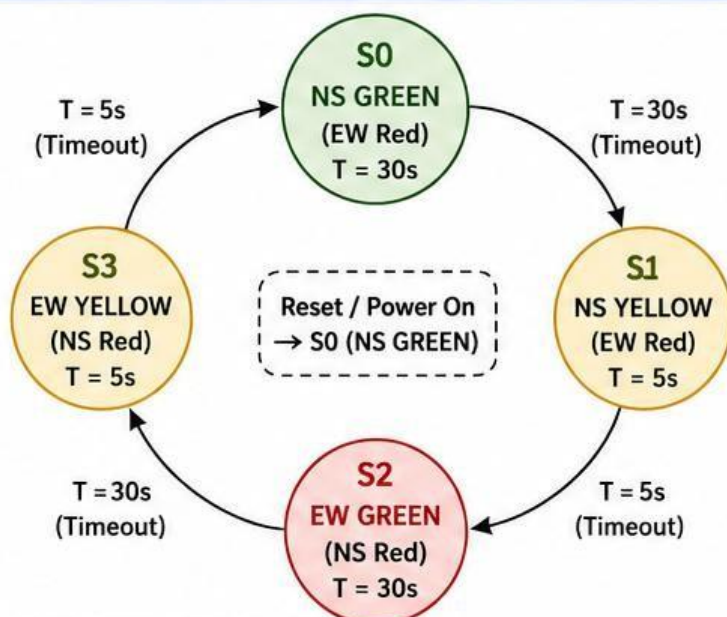
Software required: Xilinx Vivado

Hardware required: Nexys4 DDR FPGA Development Board

Theory:

Block diagram:

1. FSM STATE DIAGRAM – TRAFFIC LIGHT CONTROLLER



Note: NS = North–South , EW = East–West

State Table:

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2. STATE TABLE

Present State	Present State Bits		Next State (Timeout)	Next State Bits		Outputs (Lights)			
	Q1	Q0		Q1 ⁺	Q0 ⁺	NS Green	NS Yellow	EW Green	EW Yellow
S0 (NS GREEN)	0	0	S1	0	1	1	0	0	0
S1 (NS YELLOW)	0	1	S2	1	0	0	1	0	0
S2 (EW GREEN)	1	0	S3	1	1	0	0	1	0
S3 (EW YELLOW)	1	1	S0	0	0	0	0	0	1

Boolean Expression:

3. BOOLEAN EQUATIONS

Next State Equations

$$Q_1^+ = Q_1 \oplus Q_0$$

$$Q_0^+ = \overline{Q_1}$$

Output Equations

$$\text{NS_Green (G_N)} = \overline{Q_1} \overline{Q_0}$$

$$\text{NS_Yellow (Y_N)} = \overline{Q_1} Q_0$$

$$\text{EW_Green (G_E)} = Q_1 \overline{Q_0}$$

$$\text{EW_Yellow (Y_E)} = Q_1 Q_0$$

$$\oplus = \text{XOR} \quad , \quad \overline{A} = \text{NOT A}$$



4. STATE/OUTPUT LEGEND

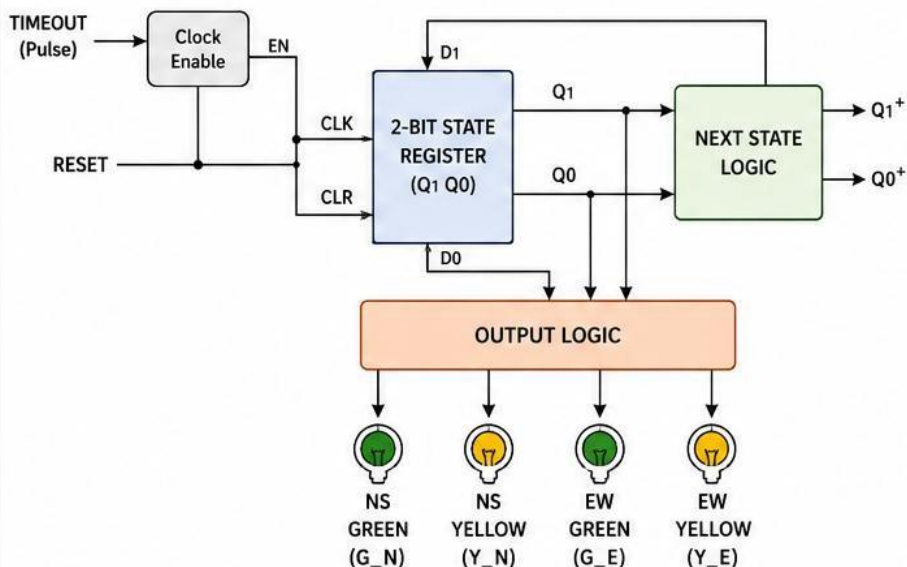
State	Q ₁	Q ₀	Traffic Light Status
S0	0	0	NS Green EW Red
S1	0	1	NS Yellow EW Red
S2	1	0	EW Green NS Red
S3	1	1	EW Yellow NS Red

Remarks:

- Timer generates a timeout pulse.
- On each timeout, the FSM moves to next state.
- Reset brings the system to S0 (NS GREEN).

Circuit Diagram:

5. CIRCUIT DIAGRAM



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Verilog Code:

```
`timescale 1ns / 1ps
```

```
module TLC_fsm #(
    parameter CLK_FREQ_HZ = 32'd100000000,
    parameter GREEN_TIME = 32'd30,
    parameter YELLOW_TIME = 32'd5
)(
    input wire CLK100MHZ,
    input wire BTNC,

    output reg NS_RED,
    output reg NS_YELLOW,
    output reg NS_GREEN,

    output reg EW_RED,
    output reg EW_YELLOW,
    output reg EW_GREEN
);
// State definitions
localparam S0 = 2'b00;
localparam S1 = 2'b01;
localparam S2 = 2'b10;
localparam S3 = 2'b11;

reg [1:0] state;
reg [1:0] next_state;

reg [31:0] counter;

// State transition
always @(*) begin

    case (state)

        // S0: NS GREEN, EW RED
        S0: begin
            if (counter >= (CLK_FREQ_HZ * GREEN_TIME) - 1)
                next_state = S1;
            else
                next_state = S0;
            end

        // S1: NS YELLOW, EW RED
        S1: begin
            if (counter >= (CLK_FREQ_HZ * YELLOW_TIME) - 1)
                next_state = S2;
            else
                next_state = S1;
            end

        // S2: NS RED, EW GREEN
```

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```

S2: begin
    if (counter >= (CLK_FREQ_HZ * GREEN_TIME) - 1)
        next_state = S3;
    else
        next_state = S2;
    end

// S3: NS RED, EW YELLOW
S3: begin
    if (counter >= (CLK_FREQ_HZ * YELLOW_TIME) - 1)
        next_state = S0;
    else
        next_state = S3;
    end

default:
    next_state = S0;

endcase

end

// State register and counter
always @(posedge CLK100MHZ or posedge BTNC) begin

    if (BTNC) begin

        state <= S0;
        counter <= 32'd0;

    end

    else begin

        if (state != next_state) begin
            state <= next_state;
            counter <= 32'd0;
        end

        else begin
            counter <= counter + 1'b1;
        end

    end

end

end

// Output logic

always @(*) begin

    // Default: all lights OFF
    NS_RED    = 1'b0;
    NS_YELLOW = 1'b0;
    NS_GREEN  = 1'b0;

```

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```
EW_RED   = 1'b0;
EW_YELLOW = 1'b0;
EW_GREEN = 1'b0;
```

```
case (state)
```

```
  S0: begin
        NS_GREEN = 1'b1;
        EW_RED   = 1'b1;
    end
```

```
  S1: begin
        NS_YELLOW = 1'b1;
        EW_RED   = 1'b1;
    end
```

```
  S2: begin
        NS_RED   = 1'b1;
        EW_GREEN = 1'b1;
    end
```

```
  S3: begin
        NS_RED   = 1'b1;
        EW_YELLOW = 1'b1;
    end
```

```
  default: begin
        NS_GREEN = 1'b1;
        EW_RED   = 1'b1;
    end
```

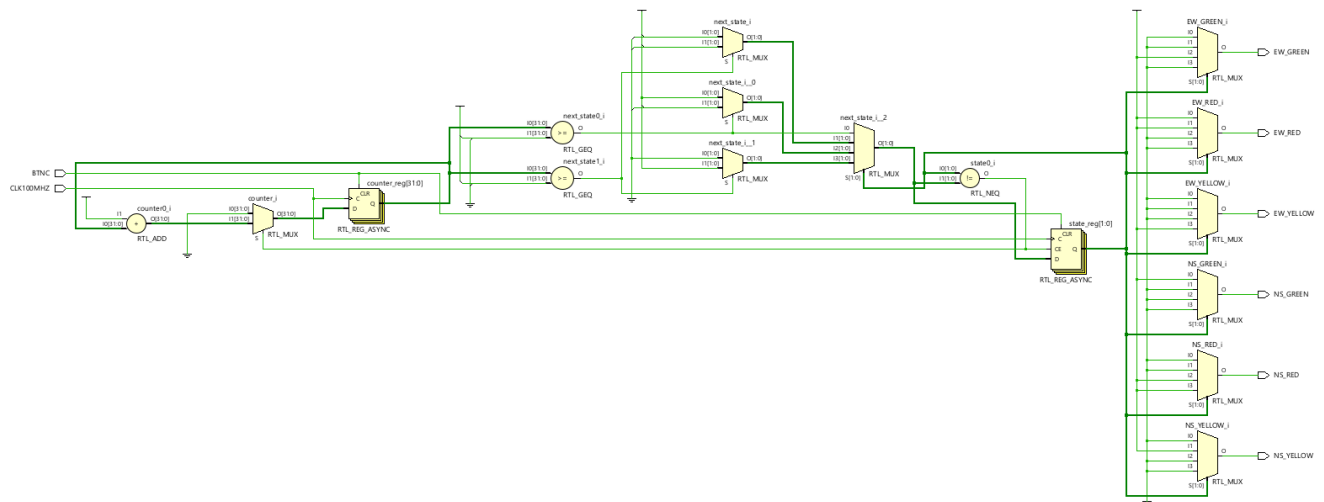
```
endcase
```

```
end
```

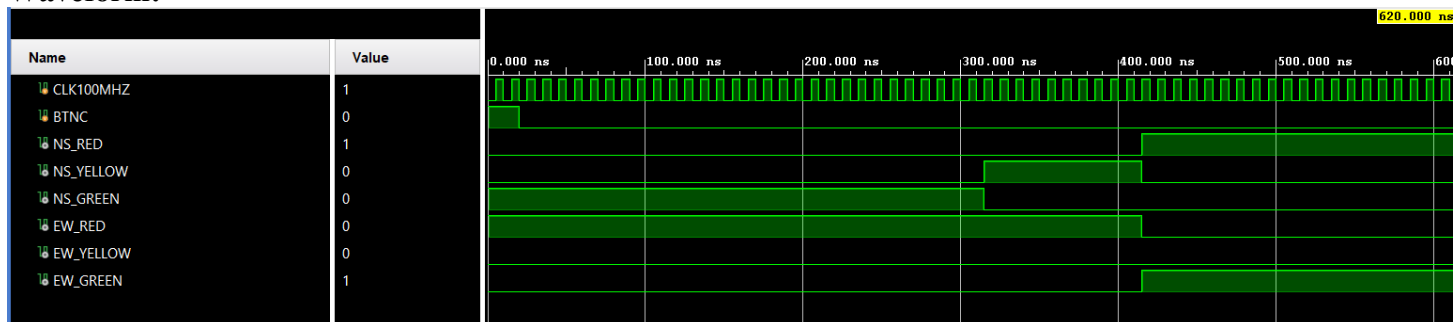
```
endmodule
```

Simulation Results:

Schematic design:

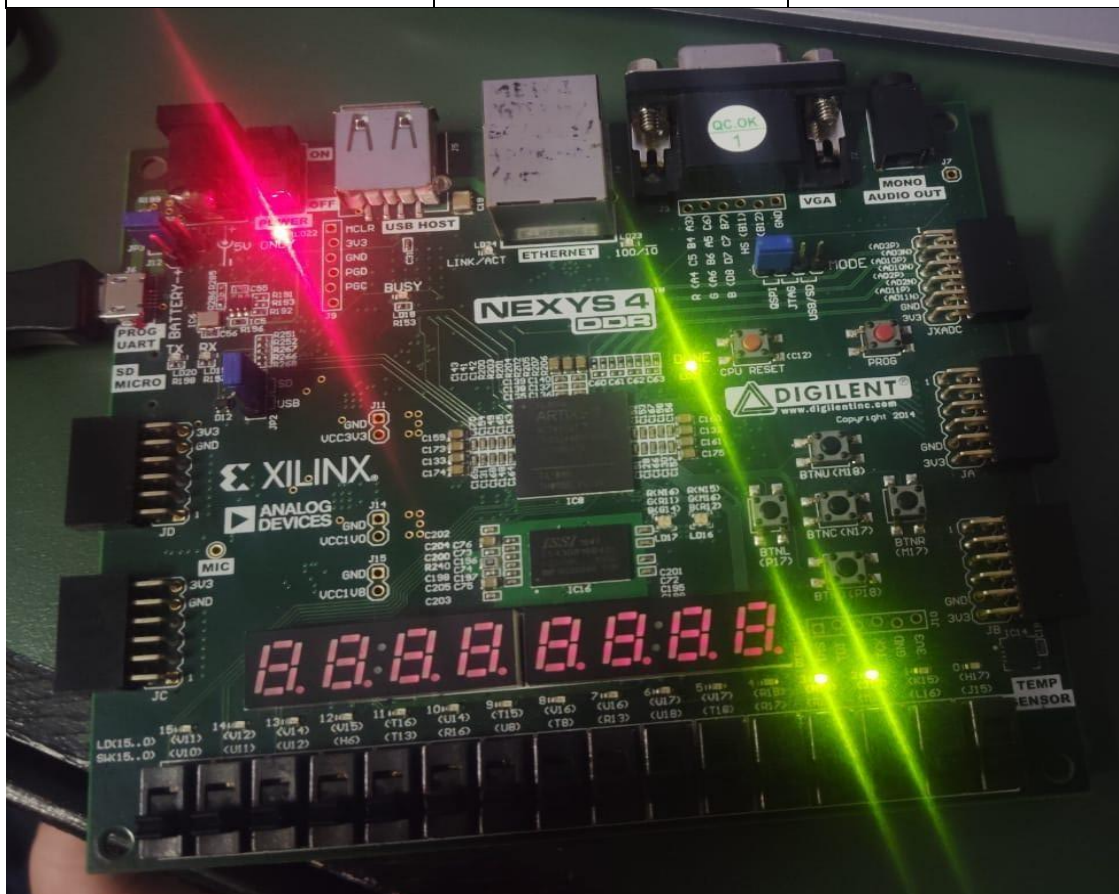


Waveform:



FPGA Implementation:

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Conclusion:

The intelligent traffic control system was successfully designed using an FSM-based approach and implemented on the Nexys4 DDR FPGA. The state transitions and traffic-light control logic were tested and verified through simulation and hardware implementation. The results demonstrated safe, synchronized, and reliable operation of the traffic signals.

Post Lab Exercise:

1. Why is an FSM suitable for designing an intelligent traffic control system?

An FSM controls traffic lights by moving through predefined states in a fixed sequence. It ensures

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predictable, safe, and reliable operation

2. How does sensor-based traffic detection improve traffic signal efficiency?

Sensors detect vehicle presence and traffic density, allowing signal timings to adjust dynamically. This reduces waiting time and improves traffic flow.

3. Why should conflicting traffic directions never receive a green signal simultaneously?

Simultaneous green signals for conflicting directions can cause accidents and traffic collisions. The FSM prevents this by allowing only safe state transitions.

4. How can adding emergency vehicle detection improve the traffic controller?

The system detects approaching emergency vehicles and gives them priority by turning the traffic signal green in the required direction. This helps emergency vehicles pass through intersections more quickly and safely.

5. Why is a reset state necessary in a traffic light FSM?

A reset state initializes the controller to a known safe condition after power-up or faults. This ensures correct and predictable operation.

6. How does state minimization benefit a traffic control FSM?

State minimization reduces the number of required states, making the design simpler and more efficient. It also lowers hardware usage and improves performance.

7. Why should pedestrian crossing requests be included in the FSM design?

Pedestrian requests are handled by assigning dedicated signal states to provide a safe crossing period. This improves pedestrian safety while also helping maintain smooth and organized traffic flow.

8. How does increasing the number of traffic lanes affect FSM complexity?

More lanes require additional signal combinations and control logic, increasing the number of states and transitions. This makes the FSM more complex.

9. Why is timing synchronization important in traffic signal control?

Proper timing synchronization ensures smooth and orderly transitions between different traffic signal states while preventing conflicting signals. This helps improve overall road safety and allows traffic to flow more efficiently.

10. How can adaptive green signal timing reduce traffic congestion?

Adaptive timing adjusts the duration of the green signal according to real-time traffic conditions. This helps minimize unnecessary waiting times and improves the overall flow of traffic.